Question

The following operations were performed on the powdered metal $\mathbf{M}$:

1. Step 1: After heating, the powder slowly dissolves in concentrated sulfuric acid, producing a gas $\mathbf{A}$ with a pungent odor.

2. Step 2: A small amount of the solution is mixed with pyridine, and hydrogen peroxide solution is slowly added under low-temperature conditions. The solution turns blue, forming a mononuclear pentagonal pyramidal complex $\mathbf{B}$. If pyridine is replaced with bipyridine, the product undergoes only minor changes.

3. Step 3: KOH solution is added to the sulfuric acid solution of $\mathbf{M}$ until nearly neutral, resulting in a gray-green precipitate $\mathbf{C}$.

4. Step 4: Continued addition of KOH solution dissolves the precipitate, forming a colored ion $\mathbf{D}$ in the solution.

5. Step 5: Hydrogen peroxide is added to the solution, causing a color change. Cooling and crystallization yield a solid $\mathbf{E}$.

6. Step 6: An appropriate amount of solid $\mathbf{E}$ is dissolved in a relatively concentrated sulfuric acid solution, producing a red solution containing two ions $\mathbf{F}$ and $\mathbf{G}$, both of which appear red. $\mathbf{F}$ and $\mathbf{G}$ are trinuclear and tetranuclear oxoacid ions, respectively, each carrying a double negative charge.

Please select the correct option.

A. In the process of dissolving $\mathbf{M}$ in sulfuric acid, the molar ratio of the generated gas $\mathbf{A}$ to $\mathbf{M}$ is 1:2

B. When pyridine in the second step is replaced with bipyridine, the structure of $\mathbf{B}$ transforms into a mononuclear pentagonal bipyramidal complex, featuring one independent oxygen atom coordination, two peroxide ligand coordinations, and two N coordinations. The coordinating atoms in the equatorial plane consist of 2 N atoms and 3 O atoms from peroxide ligands.

C. The color of $\mathbf{D}$ to $\mathbf{E}$ changes from green to orange-yellow

D. The sum of the oxygen atoms in $\mathbf{F}$ and $\mathbf{G}$ is 25.

E. The ionic structure symmetry of $\mathbf{D}$ belongs to the same point group as $Ni(CO)_4$.

F. The electron configuration of the $\mathbf{M}$ atom's extranuclear electrons conforms to the general formula $nd^5(n+1)s^1$.

Answer

Correct Answer: E

Detailed Explanation

Based on the comprehensive analysis of the question's nature, it is determined that $\mathbf{M}$ is Cr.

Option F provides a general formula containing only valence electrons, while the description refers to the extranuclear electrons of the atom. A complete electron configuration should be written, so Option F is incorrect.

CHECKPOINT 1 PTS

The general formula given in Option F contains only valence electrons and is incorrect

The reaction equation for $\mathbf{M}$ dissolving in concentrated sulfuric acid:

$$Cr + 6H_2SO_4(conc.) \rightarrow Cr(SO_4)_3 + 3SO_2 + 6H_2O$$

CHECKPOINT 1 PTS

$$Cr + 6H_2SO_4(conc.) \rightarrow Cr(SO_4)_3 + 3SO_2 + 6H_2O$$

Second Step

Combining knowledge of elemental chemistry, the generated blue complex should be $CrO(O_2)_2(py)$.

CHECKPOINT 1 PTS

The blue complex is $CrO(O_2)_2(py)$

The structure of $\mathbf{B}$ consists of one independent oxygen atom coordinated at the apex of the cone, two bidentate peroxide ligands, and one pyridine nitrogen atom coordinated at the base of the cone.

When replaced with a bipyridine ligand, the coordination changes from one N to two N atoms. Compared to the structure of $\mathbf{B}$, the additional N forms the other apex of the pentagonal bipyramid.

Since the question states that the product undergoes only minor changes, the addition of one N directly coordinating on the opposite side to form the apex of the structure is the least altered scenario. Moreover, peroxide coordination cannot form a 90-degree angle, making it impossible for one oxygen to lie in the pentagonal plane while the other occupies the apex.

Therefore, Option B is incorrect.

CHECKPOINT 1 PTS

The structure of $\mathbf{B}$ consists of one independent oxygen atom coordinated at the apex of the cone, two bidentate peroxide ligands, and one pyridine nitrogen atom coordinated at the base of the cone.

CHECKPOINT 1 PTS

When replaced with a bipyridine ligand, the coordination changes from one N to two N atoms. Compared to the structure of $\mathbf{B}$, the additional N forms the other apex of the pentagonal bipyramid.

CHECKPOINT 1 PTS

Peroxide coordination cannot form a 90-degree angle, making it impossible for one oxygen to lie in the pentagonal plane while the other occupies the apex.

Third Step: The gray-green precipitate $\mathbf{C}$ is $Cr(OH)_3$.

CHECKPOINT 1 PTS

The gray-green precipitate $\mathbf{C}$ is $Cr(OH)_3$

Fourth Step: Upon further addition of KOH solution, the precipitate dissolves, and the colored ion $\mathbf{D}$ generated in the solution is: $Cr(OH)_4^-$, appearing green and belonging to the Td point group. The option $Ni(CO)_4$ also belongs to the Td point group, so Option E is correct.

CHECKPOINT 1 PTS

$Cr(OH)_4^-$, appearing green, belongs to the Td point group

CHECKPOINT 1 PTS

$Ni(CO)_4$ belongs to the Td point group, so Option E is correct

Fifth Step: Upon adding hydrogen peroxide to the solution, the color changes. Cooling and crystallization yield solid $\mathbf{E}$. Under alkaline conditions, the product should be: K_2CrO_4, appearing yellow.

CHECKPOINT 1 PTS

$\mathbf{E}$ is K_2CrO_4, appearing yellow

Sixth Step: $\mathbf{F}$ and $\mathbf{G}$ are tri- and tetra-nuclear oxo-chromate ions, respectively, each carrying a double negative charge. The process involves no change in oxidation state: Cr remains +6, and O remains -2. Based on valence states, we obtain:

$\mathbf{F}$: $Cr_3O_{10}^{2-}$, $\mathbf{G}$: $Cr_4O_{13}^{2-}$, with the total number of oxygen atoms summing to 23.

CHECKPOINT 1 PTS

$\mathbf{F}$: $Cr_3O_{10}^{2-}$, $\mathbf{G}$: $Cr_4O_{13}^{2-}$, with the total number of oxygen atoms summing to 23

In conclusion, the correct answer is **E**.